A Geometric and Computational Investigation of the Lonely Runner Conjecture

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Abstract

This report covers my COMP-4960 project on the Lonely Runner Conjecture, which continues my Summer 2026 USRA research with Dr. Ahmad Biniaz. We study speed sets with smallest speed $v_1 = 1$ and largest speed $v_n \leq 2n$, the open part of a question raised by Tao. When $v_n \leq n$, the first and last runners always cross at a lonely time, and our goal is to find a pattern like this that works for every speed set up to $2n$, and then to use that pattern in a proof. In the summer we found that the simplest version, a lonely time where speed 1 crosses another runner, does not always exist. This report sets out three new directions. The first is to focus only on *hard* speed sets, where simple lonely times do not exist, and to describe what these sets look like. The second is to test whether, for hard sets, the best lonely time always involves a fast runner. The third is to look at the speeds that are *missing* from the set rather than the ones in it, an idea that already settles the case where $2n+1$ is prime.

1 Introduction

COMP-4960 is a two-term undergraduate research course offered by the School of Computer Science at the University of Windsor. The course provides students with an opportunity to conduct independent research under the guidance of a faculty supervisor, explore research problems, and develop practical experience in academic research. For this course, I will be continuing my previous research under the supervision of Dr. Ahmad Biniaz, focusing on the Lonely Runner Conjecture through geometric and computational approaches.

This project builds upon the research I conducted during Summer 2026 through the Natural Sciences and Engineering Research Council of Canada (NSERC) Undergraduate Student Research Award (USRA), with additional funding provided through Dr. Biniaz's research grant [28].

1.1 The problem

The Lonely Runner Conjecture considers a group of runners moving around a circular track of length 1 at distinct constant speeds, all starting from the same position. The conjecture states that, for any group of $n+1$ runners, each runner will eventually be at least $1/(n+1)$ of the track away from every other runner.

We fix one runner at the start, so the other n runners have distinct positive whole number speeds $V = \{v_1 < v_2 < \cdots < v_n\}$. For a number x, we write $\|x\|$ for the distance from x to the nearest whole number, so the runner with speed v is at distance $\|tv\|$ from the start at time t. A time t is a *lonely time* for V if

$$\|tv\| \geq \frac{1}{n+1} \quad \text{for every } v \in V,$$

and the conjecture says every speed set has a lonely time. If we plot $\|tv\|$ against t, each runner gives a zigzag, and a lonely time is a point where every zigzag is above the line at height $1/(n+1)$.

1.2 Our case and our approach

We study speed sets with $v_1 = 1$ and $v_n \leq 2n$. Tao asked whether every such set has a lonely time [6]. If $v_1 \geq 2$ the answer is known to be yes, so $v_1 = 1$ is the open part, and it is our whole focus.

Our approach starts from a simple fact. If $v_1 = 1$ and $v_n \leq n$, the first and last runners cross at

$$t = \frac{1}{1+v_n},$$

and this is always a lonely time. The goal of the project is to find a pattern like this that holds for every speed set with $v_n \leq 2n$, and then to use that pattern in a proof. Our method for each candidate pattern is:

1. search by computer for a speed set where the pattern fails;

2. if a counterexample is found, refine the pattern and try again;

3. if no counterexample is found, try to prove the pattern.

In the summer, the first patterns we tried did not survive step 1. The idea that speed 1 always crosses another runner at a lonely time fails for $\{1,3,5,8\}$, and the idea that speed 1 or speed 2 is always involved fails for $\{1,2,5,6,7,8,11\}$. Speed 1 is always in our sets, but it does not have to be part of the pattern: at any time between $1/(n+1)$ and $n/(n+1)$ the runner with speed 1 is already far enough from the start. So we now look for patterns among the other speeds.

2 Research plan

The plan is built around three ideas, from the most direct to the most ambitious. They build on each other: Idea 1 tells us which speed sets matter, Idea 2 tests a pattern on exactly those sets, and Idea 3 looks for a proof.

2.1 Idea 1: focus on hard speed sets

Many speed sets have an easy lonely time. If V contains no multiple of some number m with $2 \le m \le n+1$, then at $t = 1/m$ every runner is at distance at least $1/m \ge 1/(n+1)$, so $t = 1/m$ is a lonely time. Any pattern only has to work for the remaining sets.

Definition 1. A speed set V is *hard* if it contains a multiple of every number m with $2 \le m \le n+1$.

This is a strong condition, and it already tells us some of what a hard set must look like:

- The only multiple of $n+1$ that is at most $2n$ is $n+1$ itself, so every hard set contains $n+1$.
- If $2n/3 < m \le n$, the only multiples of m that are at most $2n$ are m and $2m$, so every hard set contains m or $2m$.

What to do. List the hard sets for small n and describe them. Our guess is that they look like $\{1,2,\ldots,n\}$ with a few speeds doubled. A clear description would tell a proof exactly which sets it has to handle.

2.2 Idea 2: the size of the best pair on hard sets

For each speed set, the best lonely time is reached where two runners cross (or at the peak of one runner). In earlier work we tested whether this best pair always contains a speed larger than $n/2$. The rule fails for some sets, but every failure we found comes from an easy set. On hard sets it has not failed for any $n \le 13$.

> **Question.** If V is hard, does the best lonely time always come from a pair of runners that contains a speed larger than $n/2$?

What to do. Follow the method from the introduction: search hard sets for larger n for a counterexample. If none appears, look at why the fast runner is always involved and try to prove it.

2.3 Idea 3: look at the missing speeds

A speed set uses n of the speeds from 1 to $2n$, so n speeds are missing. Instead of asking which runners in V cross, we can ask which speeds are *not* in V. The key observation is the following. Suppose $2 \leq b \leq n$ and b has no common factor with $2n+1$. If neither b nor $2n+1-b$ is in V, then there is a lonely time with denominator $2n+1$. This is the same mechanism used by Bohman and Peng [7].

When $2n+1$ is prime, every b has no common factor with $2n+1$, and a simple counting argument shows that a suitable pair of missing speeds always exists. This proves our case for every n with $2n+1$ prime. The same argument appears in a recent preprint by Bonfioli [27], so it is a known starting point rather than a new result.

What to do. Write out the prime case in full, then study the smallest n where $2n+1$ is not prime ($n = 4, 7, 10, 12, 13$) to see which lonely times are used instead. A pattern there could extend the proof to more values of n.

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